

Total No. of Questions : 10]

SEAT No. :

P3534

[5560]-186

[Total No. of Pages : 2

T.E. (Computer Engineering)

**PRINCIPLES OF CONCURRENT AND DISTRIBUTED
PROGRAMMING**

(2012 Course) (Semester-II) (310249)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Question 1 or 2, 3 or 4, 5 or 6, 7 or 8 and 9 or 10.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*

Q1) a) What are the different computational Models? Explain in detail. **[6]**

b) Write a program in LISP to find factorial of a given number. **[4]**

OR

Q2) a) Explain inter process communication. **[6]**

b) Explain how to count task dependency. **[4]**

Q3) a) Write a note on Feng's classification. **[6]**

b) Explain various types of parallelism. **[4]**

OR

Q4) a) Explain: **[6]**

- General purpose computer architecture.
- Special purpose computer architecture.

b) Compare GPU and CPU. **[4]**

P.T.O.

- Q5) a)** What are the major issues of designing a Distributed OS? [10]
b) List and explain any two transparencies of a distributed system with a suitable example. [8]

OR

- Q6) a)** What is distributed computing system? Explain tightly and loosely coupled system with neat diagram. [10]
b) Explain the processor pool model along with advantages and disadvantages of it? [8]

- Q7) a)** Explain Domain0 and DomainU in Xen? [8]
b) What is memory and MMU virtualization? [4]
c) What is Hardware virtualization? [4]

OR

- Q8) a)** What is need of Virtualization? Explain types of virtualization. [8]
b) What is Kernel-level virtualization? [4]
c) What are the advantages of virtualization? Explain. [4]

- Q9) a)** Explain the concept of mobile computing with respect to the following points: [6]
i) Mobile computing classification.
ii) Advantages.
iii) Security issues before mobile computing.
b) Write short notes on: [6]
• CUDA grids
• CUDA Kernels
c) Write a CUDA program for addition of two matrices. [4]

OR

- Q10) a)** Explain threads in CUDA. Also explain problem decomposition. [6]
b) Explain multi-GPU model in single-node systems in CUDA. [6]
c) Explain CUDA Task Execution Model. [4]

